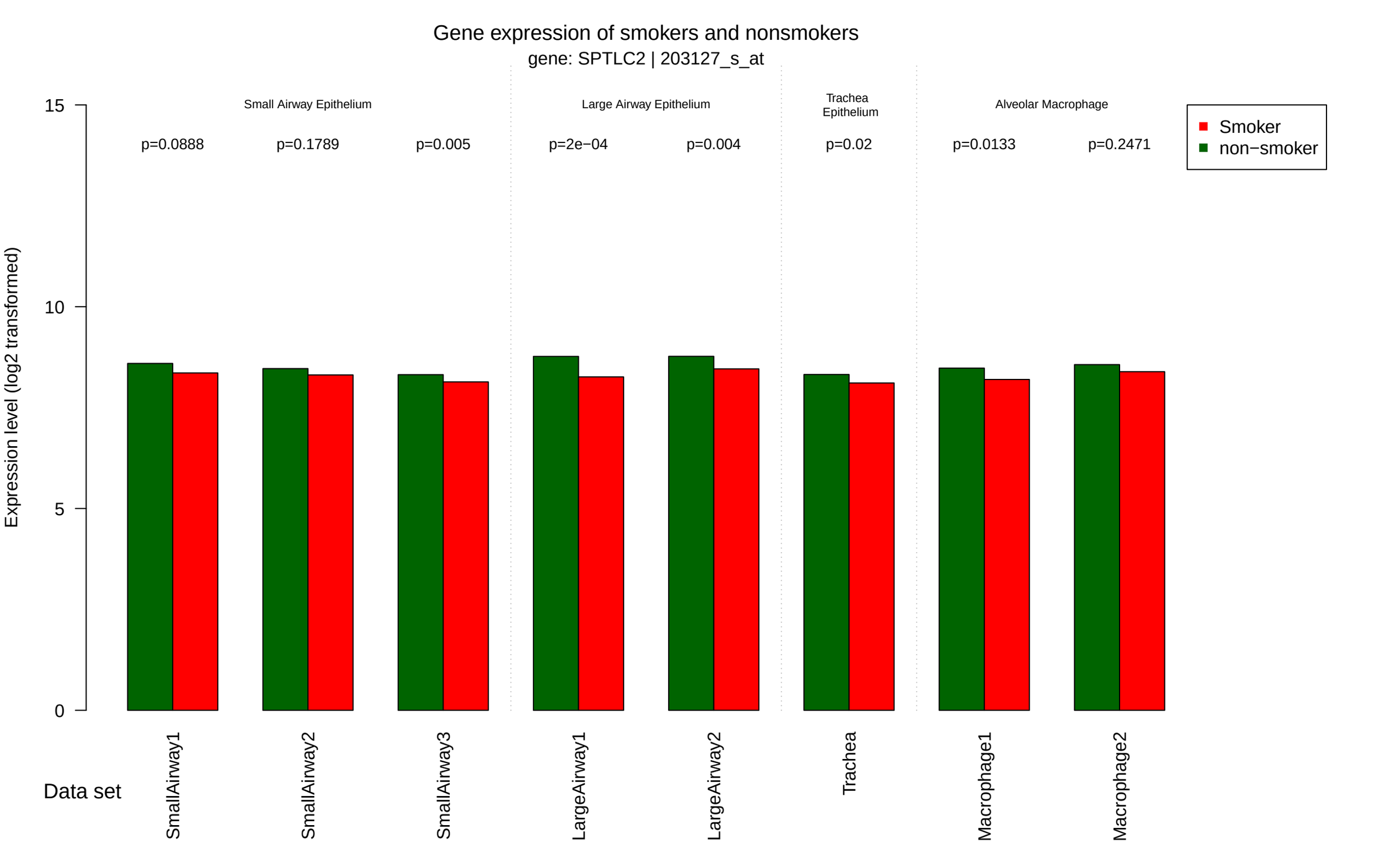
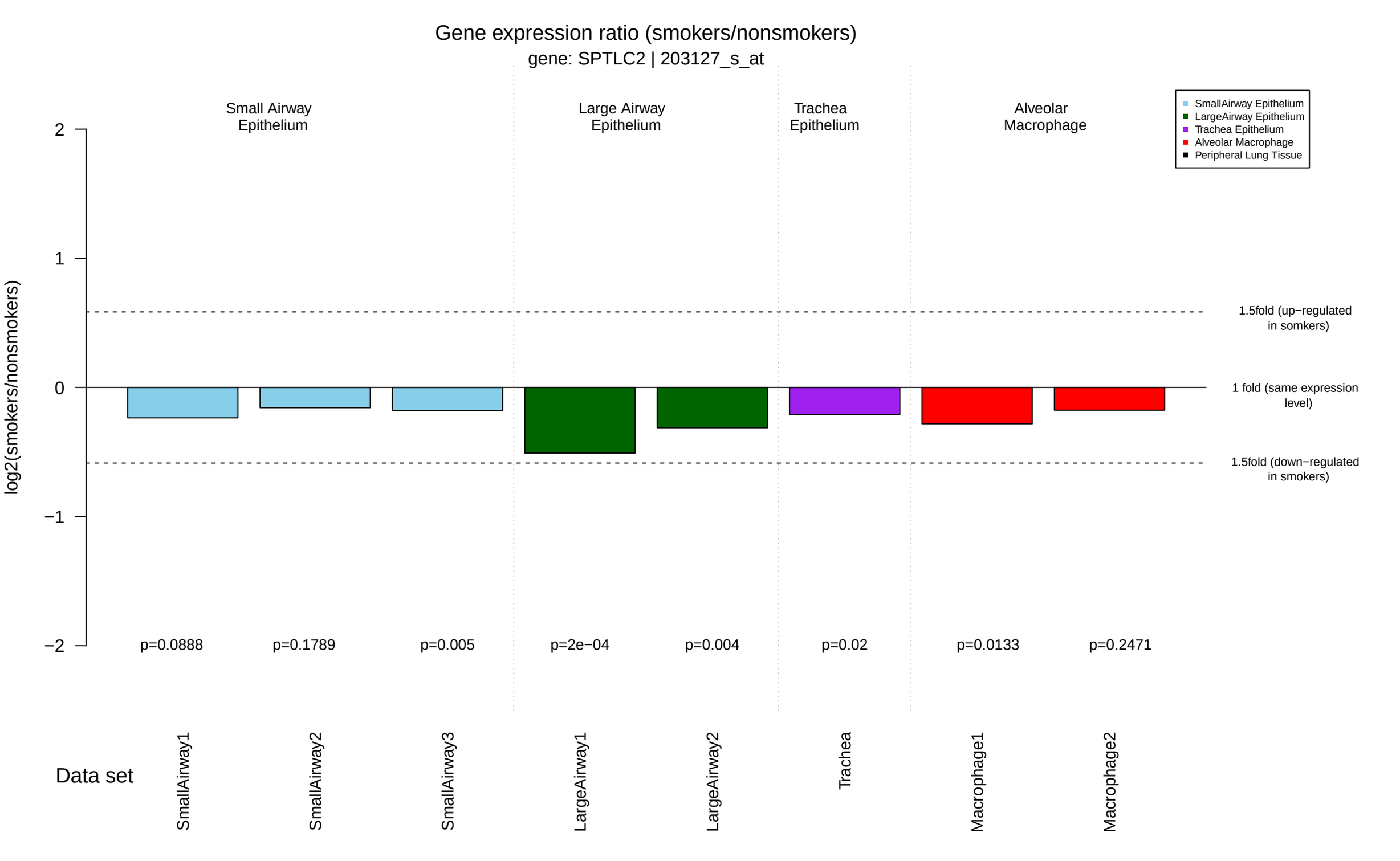
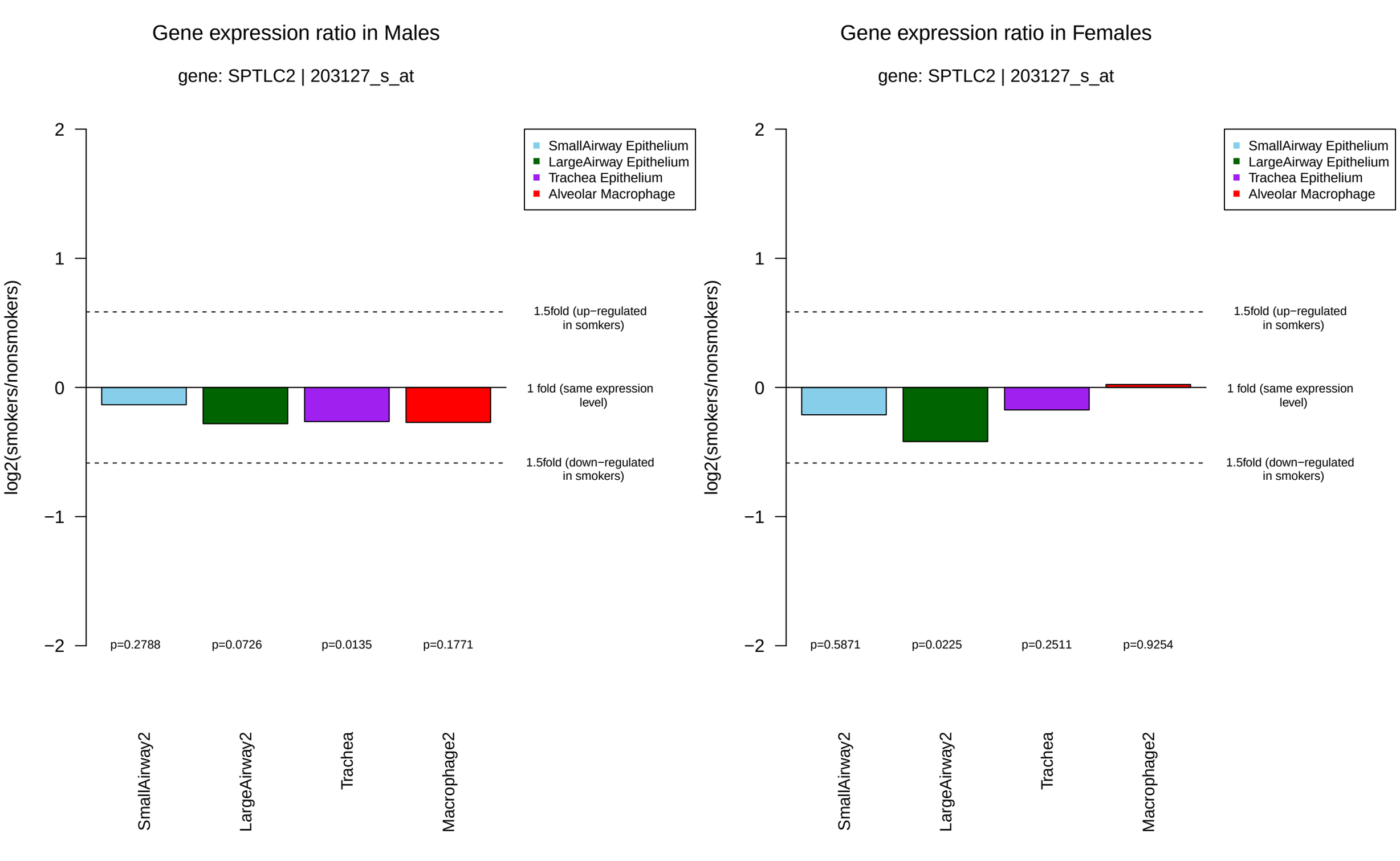
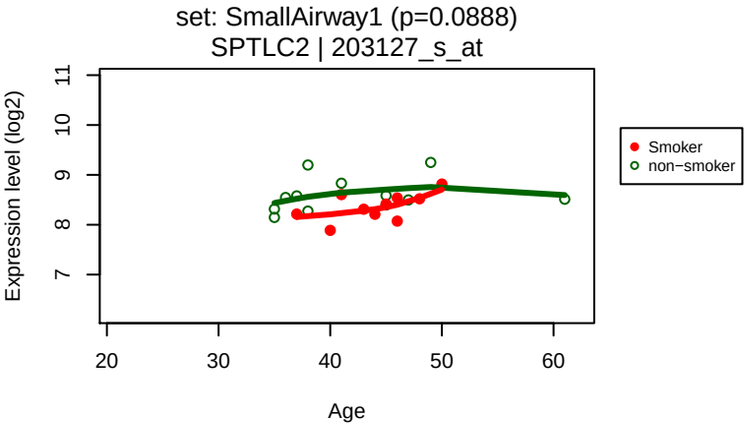
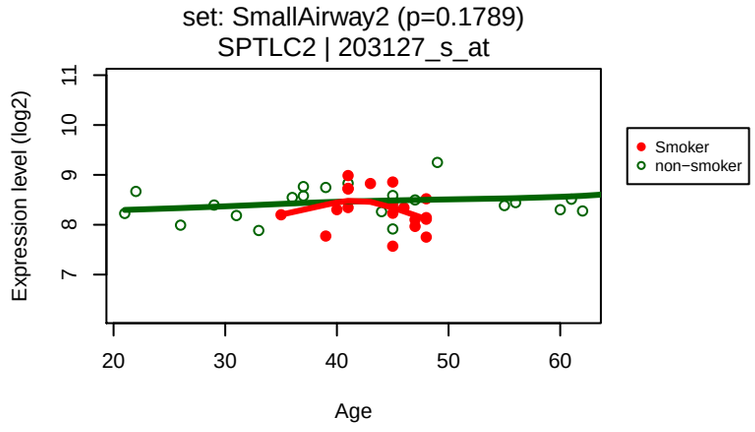
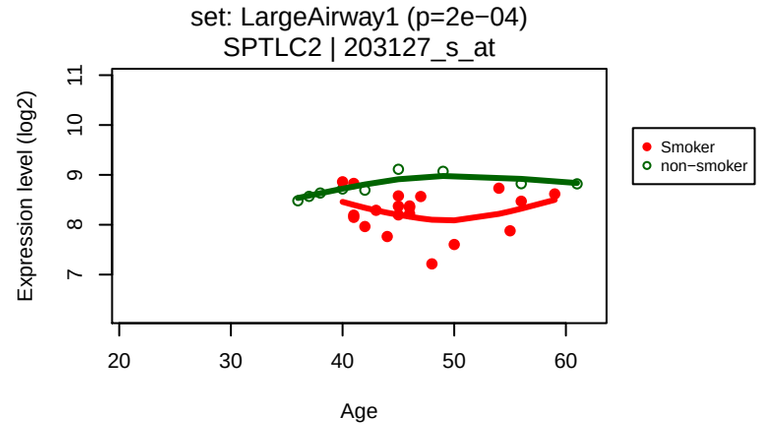

Gene expression of smokers and nonsmokers
gene: SPTLC2 | 203127_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0888
p=0.1789
p=0.005
p=2e−04
p=0.004
p=0.02
p=0.0133
p=0.2471
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

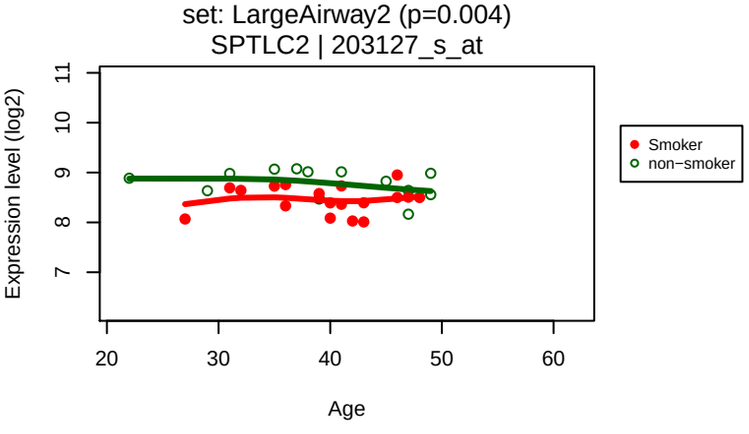

Gene expression ratio (smokers/nonsmokers)
gene: SPTLC2 | 203127_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0888
p=0.1789
p=0.005
p=2e−04
p=0.004
p=0.02
p=0.0133
p=0.2471
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

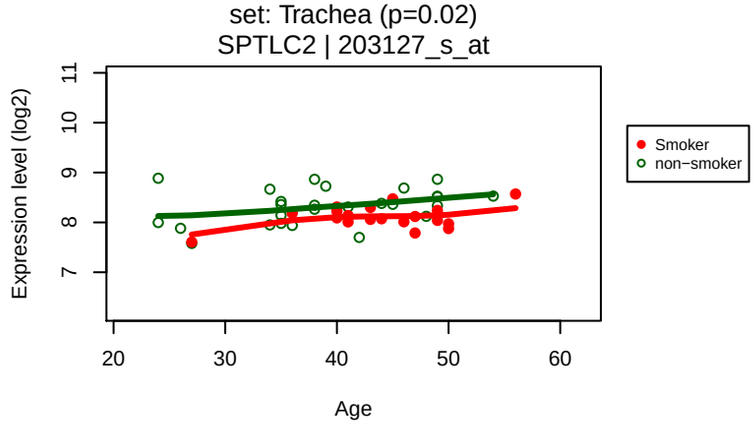

Gene expression ratio in Males
gene: SPTLC2 | 203127_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2788
p=0.0726
p=0.0135
p=0.1771
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPTLC2 | 203127_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5871
p=0.0225
p=0.2511
p=0.9254
SmallAirway2
LargeAirway2
Trachea
Macrophage2

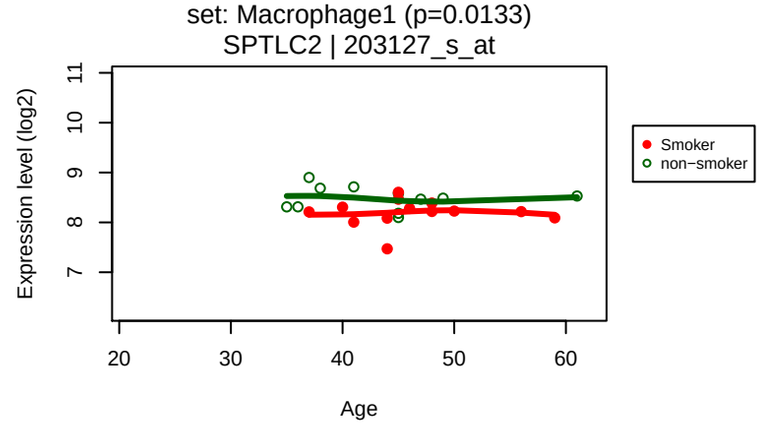

set: SmallAirway1 (p=0.0888)
SPTLC2 | 203127_s_at
Expression level (log2)
Age
Smoker
non−smoker

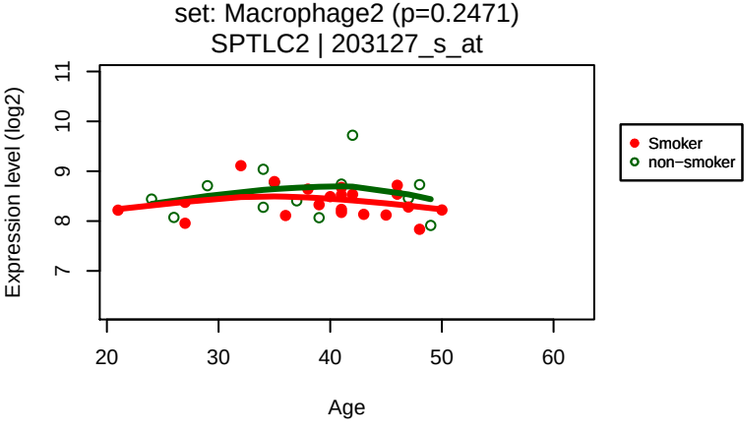

set: SmallAirway2 (p=0.1789)
SPTLC2 | 203127_s_at
Expression level (log2)
Age
Smoker
non−smoker

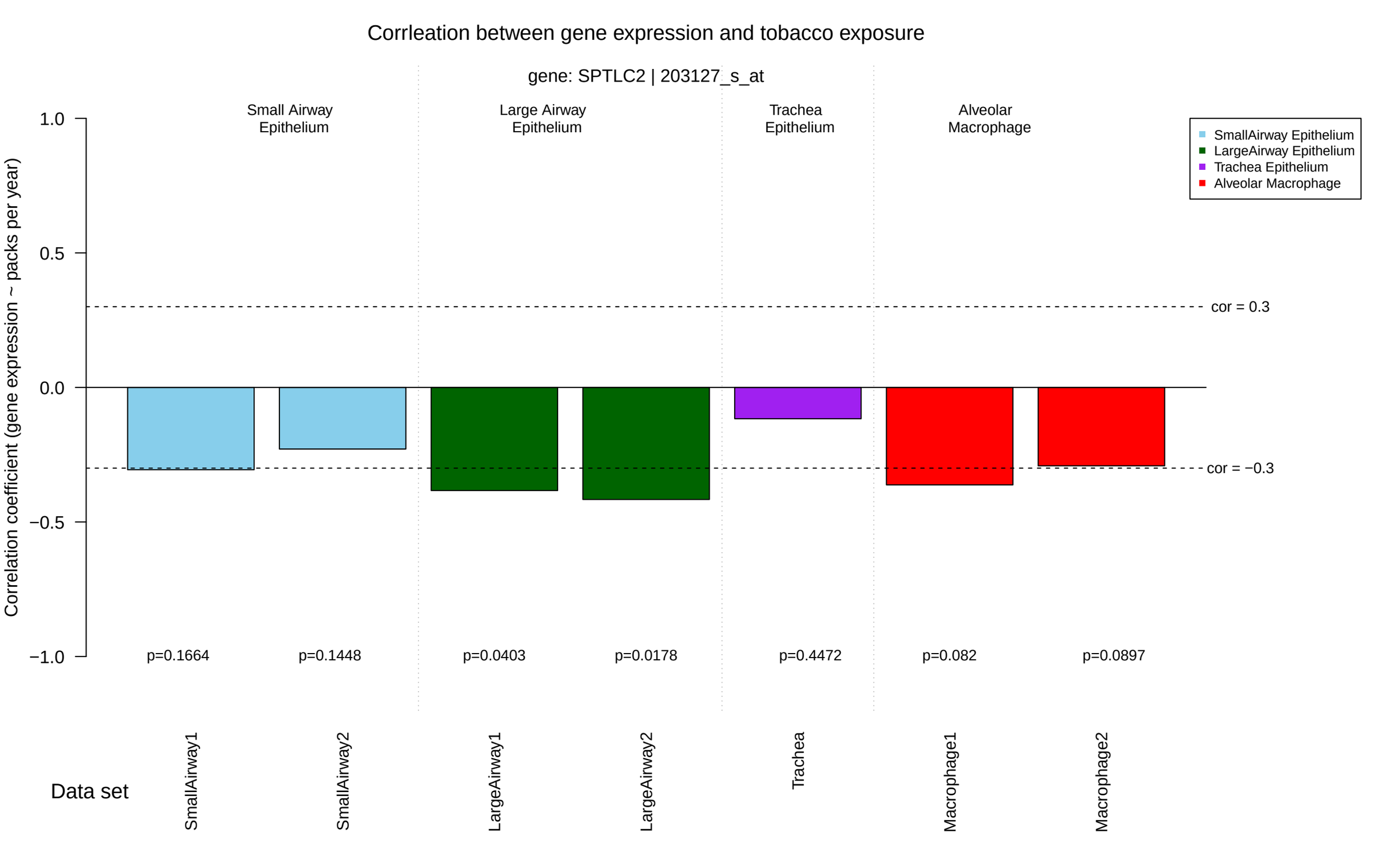

set: LargeAirway1 (p=2e−04)
SPTLC2 | 203127_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.004)
SPTLC2 | 203127_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.02)
SPTLC2 | 203127_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0133)
SPTLC2 | 203127_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2471)
SPTLC2 | 203127_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPTLC2 | 203127_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1664
p=0.1448
p=0.0403
p=0.0178
p=0.4472
p=0.082
p=0.0897
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2